KenganSocial: Architecture & Cost Analysis

This document provides a detailed overview of the KenganSocial web application architecture, all integrated API connections, and an estimated cost breakdown for running the platform with 50 users on the basic plan.

1. System Architecture Overview

The system utilizes a modern hybrid microservices architecture to balance UI responsiveness with heavy video-processing demands.

Frontend (Client)

- **Framework:** React 18 built with Vite
- **State Management:** React Context API (`AuthContext`, `WorkspaceContext`)
- **Styling:** Custom Vanilla CSS for a premium dark aesthetic
- **Key Features:** CapCut-style local video editor timeline, Drag-and-drop mechanics, Dashboard routing via `react-router-dom`

Primary Backend (Node.js)

- **Framework:** Node.js with Express (Running on Port 5000)
- **Role:** Handles all core application logic including user management, workspace administration, social media OAuth connections, analytics aggregation, and billing.
- **Security:** Rate-limiting, CORS, Helmet, and token verification via Firebase Admin.

Video Processing Microservice (Python)

- **Framework:** Python FastAPI (Running on Port 8000)
- **Role:** Handles heavy, non-blocking asynchronous tasks such as FFmpeg video rendering, Whisper transcriptions for captions, audio mixing, and Auphonic integrations.

Database & File Storage

- **Database:** Firebase Firestore (Serverless NoSQL) for storing users, content metadata, connections, and analytics.
- **Authentication:** Firebase Authentication (Email/Password & Google OAuth).
- **Storage:** Firebase Cloud Storage for raw video uploads, user avatars, and exported media.

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2. Total API Connections

The application relies on several distinct internal and external APIs to function smoothly from signup to social media export.

API Category	Service Used	Purpose
Authentication & Auth	Google Identity / Firebase Auth	User signup, login, Google OAuth SSO, secure JWT validation.
Database & Storage	Firebase Admin SDK	Server-to-database communication for reading/writing Firestore documents and secure bucket access for video uploads.
Payment Processing	Stripe API	Subscription management, checkout sessions, and handling webhooks for the "Basic" and "Pro" plans.
Artificial Intelligence	OpenAI API (GPT-4o)	Generating social media captions, analyzing content, and AI autocomplete.
Artificial Intelligence	Google Gemini API	Fallback/Advanced AI generation for content analysis and metadata generation.
Social Media Auth & Post	YouTube Data API v3	Connecting channels, fetching analytics, publishing shorts/videos.
Social Media Auth & Post	Meta Graph API	Connecting Instagram and Facebook Pages, retrieving engagement analytics, and posting reels.
Social Media Auth & Post	TikTok for Developers API	Managing TikTok account connections and direct video publishing.
Social Media Auth & Post	X (Twitter) API & LinkedIn API	Text and media post syndication and analytics.
Audio Processing	Auphonic API (Optional integration)	Broadcast-grade audio mastering and normalization during the video render pipeline.

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3. Expected Costs for 50 Basic Users

For 50 active users on a "Basic Plan", the infrastructure footprint is relatively light, with the only significant variable being video rendering compute.

A. Free/Negligible Costs

- **Firebase Authentication:** Free (100% covered by the Spark plan).
- **Firebase Firestore:** 50 users will easily fall under the generous free tier limits (50k reads/day, 20k writes/day). **$0/month**.
- **Firebase Storage:** The free tier includes 5GB. 50 users processing short-form clips might exceed this slightly depending on retention policies (e.g., deleting raw files after render). Assuming careful cleanup, costs will be heavily mitigated (Storage is $0.026/GB beyond the free tier). Estimated at **$0 - $2/month**.
- **Stripe:** Pay-per-transaction (2.9% + 30¢). No fixed monthly server costs.

B. Fixed Infrastructure Costs

- **Node.js Backend Server:** A standard VPS (e.g., DigitalOcean Basic Droplet, Render, or Railway) requires about 1GB - 2GB RAM to smoothly handle API traffic for 50 concurrent sessions. Estimated at **$10 - $15/month**.
- **Python Video Microservice:** Video rendering via FFmpeg is CPU-intensive. A dedicated CPU-optimized instance (e.g., DigitalOcean CPU-Optimized or Hetzner CPX21/31) is highly recommended so that rendering doesn't freeze the main app. Estimated at **$20 - $35/month**.

C. Variable Usage Costs

- **OpenAI / Gemini API:** LLM generation for text/captions for 50 users is extremely cheap using current models (e.g., GPT-4o-mini). Assuming 20 API calls per user

per day, this maps to roughly **$5 - $10/month**.

Estimated Total Operational Cost

For 50 active users, expect your infrastructure and API costs to reliably range between **$35.00 to $62.00 per month**.

(Note: These costs scale very linearly. The bulk of the cost goes toward the server running the Python video rendering engine. If users render long 10-minute videos instead of 30-second shorts, the compute requirements for the Python server will increase).